

Researchers are investigating new materials, such as diamonds and graphene, for more robust qubits. Advances in miniaturisation, error correction, and quantum networking will help realise scalable and commercially viable quantum computers. The integration of artificial intelligence in quantum control systems, the rise of quantum cloud services, and the search for room-temperature qubits are shaping the next wave of breakthroughs.

Quantum hardware development: What lies ahead

The current era of Noisy Intermediate-Scale Quantum (NISQ) computing has proven that quantum processors can perform tasks beyond the reach of classical computers. However, the true promise of quantum computing—solving intractable problems in medicine, materials science, and finance—awaits the development of large-scale, fault-tolerant machines. The next 5 to 15 years will be defined by a critical transition, focusing on the key innovations required to bridge this gap.

The path to massive scalability:

Scaling from hundreds of qubits to millions is the primary engineering challenge. The future lies not in simply building larger monolithic chips, but in rethinking the fundamental architecture of a quantum processor.

- **Modular architectures (chiplets):** Inspired by modern classical CPUs, the future is modular. Quantum processors will be constructed from smaller, high-yield ‘quantum chiplets’ that are interconnected. This will allow manufacturers to select only the highest quality chiplets and combine them, dramatically improving the yield and cost-effectiveness of building a large-scale system.
- **Advanced 3D integration:** To manage the immense complexity of wiring and control, quantum hardware will move into the third

dimension. This involves stacking layers of qubit arrays on top of classical control and readout layers (heterogeneous integration), connected by vertical links. This approach drastically increases qubit density and reduces signal latency.

- **Architectural convergence:** While different qubit modalities (superconducting, silicon spin, trapped ions) currently have distinct advantages, we will see a convergence of ideas. For example, the unparalleled manufacturing scalability of silicon will influence superconducting qubit design, while the high connectivity of trapped-ion systems will inspire new routing techniques in solid-state architectures.

The pursuit of qubit quality and coherence:

A million poor-quality qubits are useless. The relentless pursuit of higher quality—longer coherence times and more precise operations—is paramount. The focus will shift from demonstrating basic functionality to achieving industrial-grade precision.

- **Materials science breakthroughs:** The quality of a qubit is determined by its environment. Future progress will depend on breakthroughs in materials science, such as producing isotopically pure Silicon-28 to eliminate magnetic noise for spin qubits, and engineering ‘cleaner’ superconducting materials with fewer microscopic defects that cause decoherence.
- **Surpassing 99.99% gate fidelity:** Achieving the ‘four nines’ of fidelity for two-qubit gates is a critical threshold for efficient quantum error correction. This will be achieved through a combination of AI-driven pulse shaping, which designs optimal electromagnetic pulses for gate operations, and designing qubits that are inherently less prone to errors and crosstalk.
- **Maintaining quality at scale:** Ensuring that the millionth

qubit is as good as the first is a monumental challenge. This requires developing fabrication processes with unprecedented uniformity and creating new techniques for shielding large arrays of qubits from correlated noise that can affect the entire processor at once.

The dawn of practical fault-

tolerance: Fault-tolerance is the paradigm shift that will unlock quantum computing’s full potential. This involves moving from fragile physical qubits to robust, error-corrected ‘logical qubits’.

- **Hardware-efficient QEC codes:** Early error-correction codes, like the surface code, require thousands of physical qubits to create one logical qubit. The future lies in developing more efficient codes that drastically reduce this overhead. This will make fault-tolerance achievable with far fewer qubits, bringing the timeline for a useful logical processor forward significantly.
- **Real-time decoding and feedback:** Quantum error correction is not a post-processing step; it must happen in real-time. This requires a dedicated classical co-processor, likely operating at cryogenic temperatures, that can detect an error, diagnose its cause, and apply a correction faster than the quantum state decoheres—a feedback loop operating on nanosecond timescales.
- **The first logical qubits:** Within the next 5-10 years we will see the first demonstrations of logical qubits that have a longer lifetime and perform more accurate operations than any of their constituent physical qubits. This will be the ‘Wright brothers moment’ for fault-tolerant quantum computing, proving that building a stable quantum memory is possible.

Interconnectivity and the quantum

internet: To scale beyond the limits of a single cryogenic refrigerator, quantum computers must learn to talk to each